



A project's papers, in one place.

Group papers into workspaces, read all their notes together, and ask questions across them.

Dense retrieval

3 papers · 3 notes

Papers

Notes

Chat

BERT: Pre-training of Deep Bidirectio...

Jacob Devlin, Ming-Wei Chang et al. · 2019 ...

Dense Passage Retrieval for Open-Do...

Vladimir Karpukhin, Barlas Oğuz et al. · 202...

Retrieval-Augmented Generation for ...

Patrick Lewis, Ethan Perez et al. · 2020 · N...

Swipe →

1 • CREATE A WORKSPACE

Start with a name.

Click New workspace in the library's sidebar, type a name and press Enter. Its page opens.

||| All papers

Workspaces

Contrastive learning

Question answering

Dense retrieval |

Enter



Dense retrieval

0 papers · 0 notes

Papers

Notes

Chat

+ Add papers

No papers in this workspace yet.





2 • ADD PAPERS

Pick papers from your library.

Add papers lists what isn't in the workspace yet. Tick a few and add them at once, or upload new PDFs.

Add papers

🔍 Search your library...

Attention Is All You Need

BERT: Pre-training of Deep Bidirectional Transform... ✓

Dense Passage Retrieval for Open-Domain Questi... ✓

Retrieval-Augmented Generation for Knowledge-In... ✓

Cancel

Add 3 papers





3 · REUSE PAPERS

One paper, in several workspaces.

Tick workspaces from any paper's menu. Removing it from one keeps it in your library.

Attention Is All You Need

Ashish Vaswani, Noam Shazeer et al. · 2017 · NeurIPS

Dense Passage Retrieval for Open-Domain Question An...

Vladimir Karpukhin, Barlas Oğuz et al. · 2020 · EMNLP



Contrastive learning

Dense retrieval ✓

Question answering ✓

 Add to workspace... >



4 · READ THE NOTES

Every note, from every paper.

The Notes tab gathers the notes on all its papers, grouped by paper. Click one to open it in the reader.

Papers

Notes

Chat

3 papers · 3 notes

Dense Passage Retrieval for Open-Domain Question Answering

You

p. 3

| in-batch negatives

Negatives come free.

AI

p. 5

| compared with BM25

Beats BM25 on most QA sets.

Retrieval-Augmented Generation for Knowledge-Intensive Question Answering

You

p. 4

| keep the document encoder
fixed

Only the query side is tuned.





5 · ASK THE WORKSPACE

One question, the whole workspace.

Answers cite passages from its papers, and your notes too.
Click a citation to see it in the paper.

Papers

Notes

Chat

3 papers · 3 notes

How do these papers train the retriever?

AI

DPR trains two BERT encoders with in-batch negatives [\[C1\]](#).
RAG starts from that retriever and fine-tunes only its query
encoder [\[C2\]](#)[\[N1\]](#).

N1 · You · Lewis 2020 p.4

Ask about this workspace...





Read deeper. Keep every thought. **Ask a whole project.**

PaperLab runs on your own machine. Your PDFs, notes and workspaces stay on your computer.

■ github.com/khannoussi-malek/PaperLab